MoodyClues

Final Submission

Team 3

Introduction

Introduction

There is a prevalence and rising trend in poor mental health in Singaporeans.

Almost 1 in 3 people between 15 and 35 years of age have reported “severe or extremely severe symptoms of depression, anxiety and/or stress.”

Research on journaling has found that it can help manage mental health symptoms.

But what if journals could be used to monitor the emotional well-being of those that might be vulnerable or those that are unable to communicate easily?

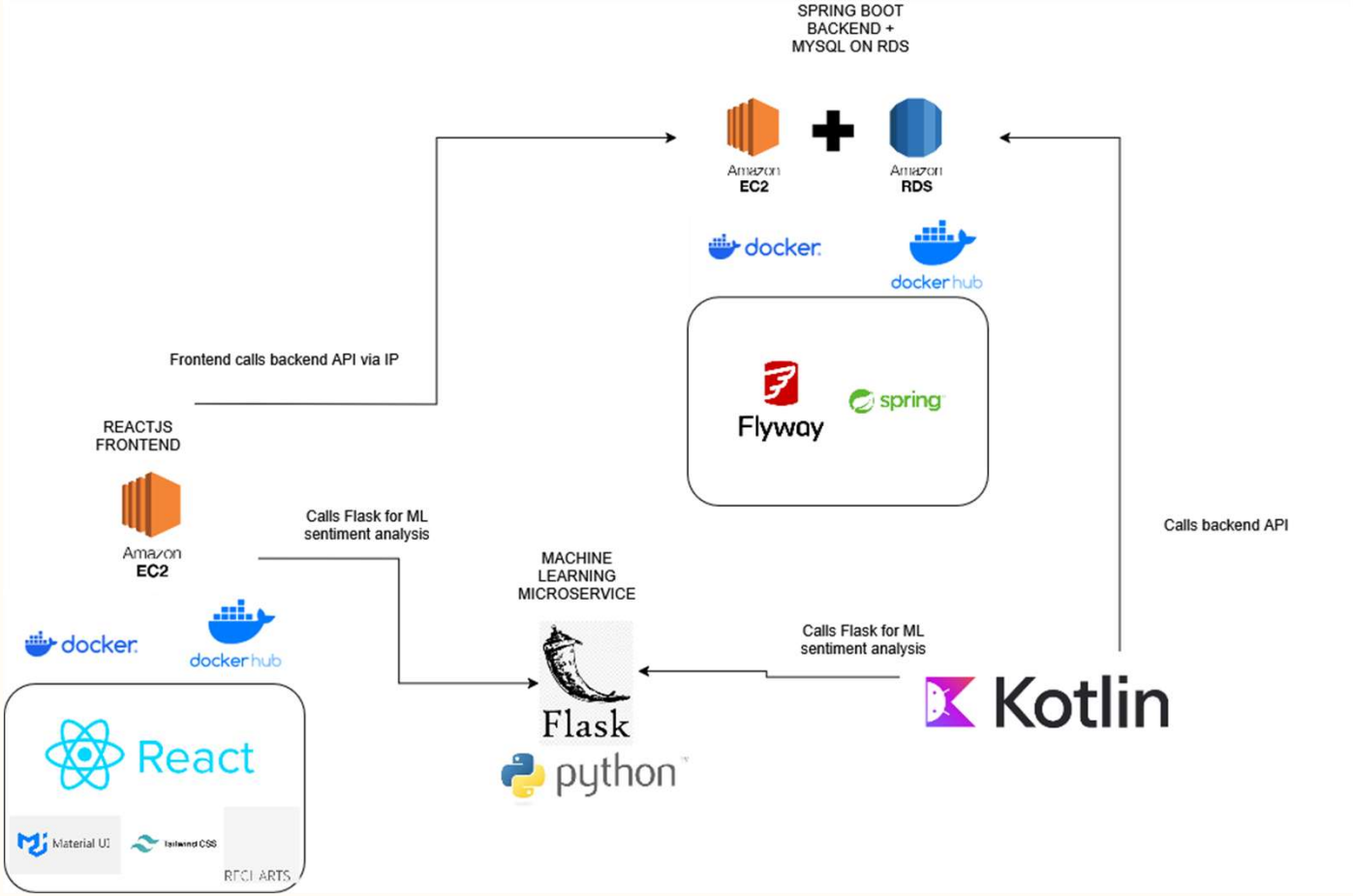
Introducing

MoodyClues

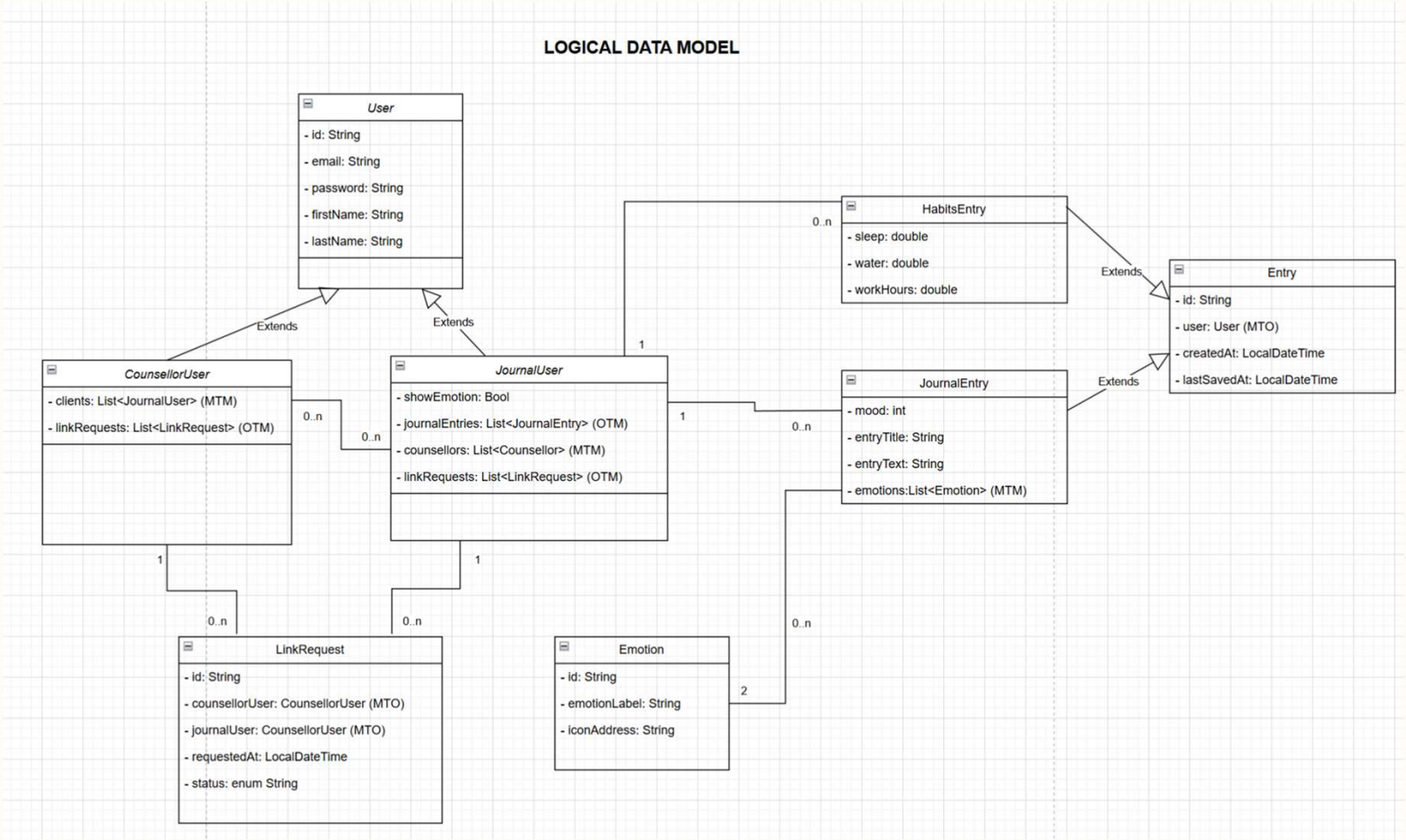
for Journaling and Emotional Well-being Monitoring

Logical Data Model and Database Schema

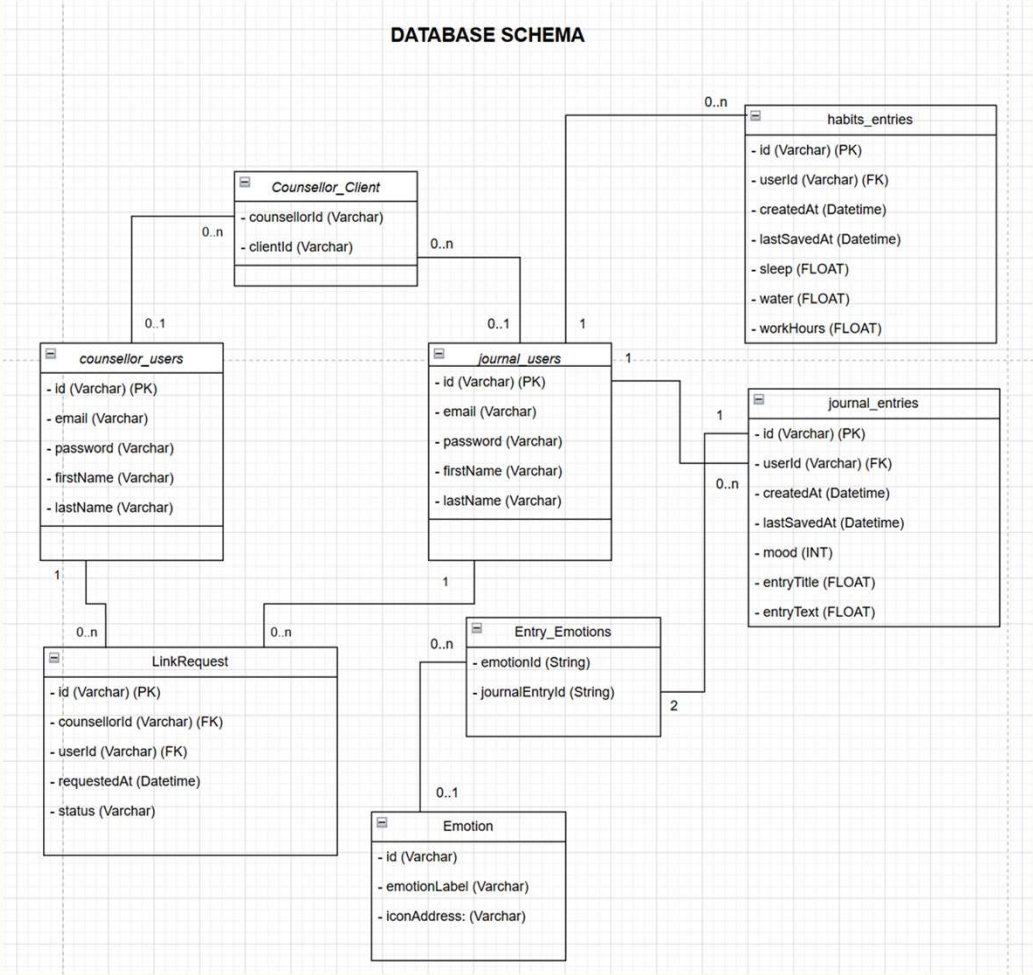
Application and Deployment Architecture



Logical Data Model



Database Schema

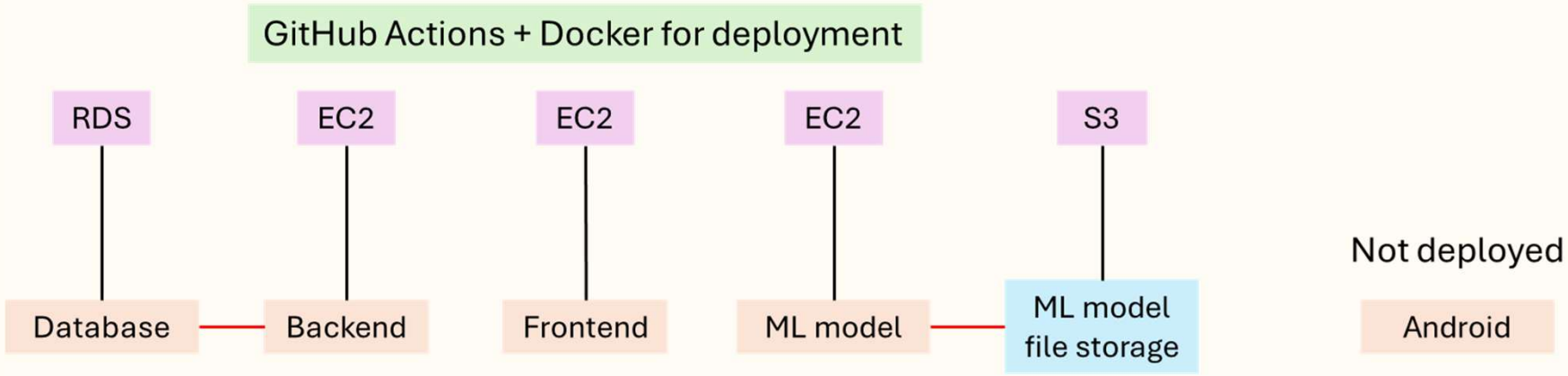


Application Main Deliverables

Feature / Deliverable	Member in-charge
- Web mock screens (All) - Spring Boot Backend (All code + architecture planning + API testing & documentation) - ReactJS (Login, Register, Home, User + Counsellor Dashboard Screens) - Backend + Database + Frontend Deployment (AWS EC2 + RDS)	Dion
- Machine learning model for sentiment analysis (Python) - ML model deployment and CI/CD pipeline implementation (FlaskAPI, AWS EC2 + S3, GitHub Actions) - ReactJS (User screens)	Hiroyo
- Android journal users' screens (Login, Register, Home,Notification,Emtionsetting) - Android journal users' screens (Login, Register, Home,Invitation) - Android call backend (All)	Ma Li
- Android Counsellor functionality	Thina
- ReactJS - Counsellor screens	Shengyi

DevSecOps Architecture

Deployment to AWS

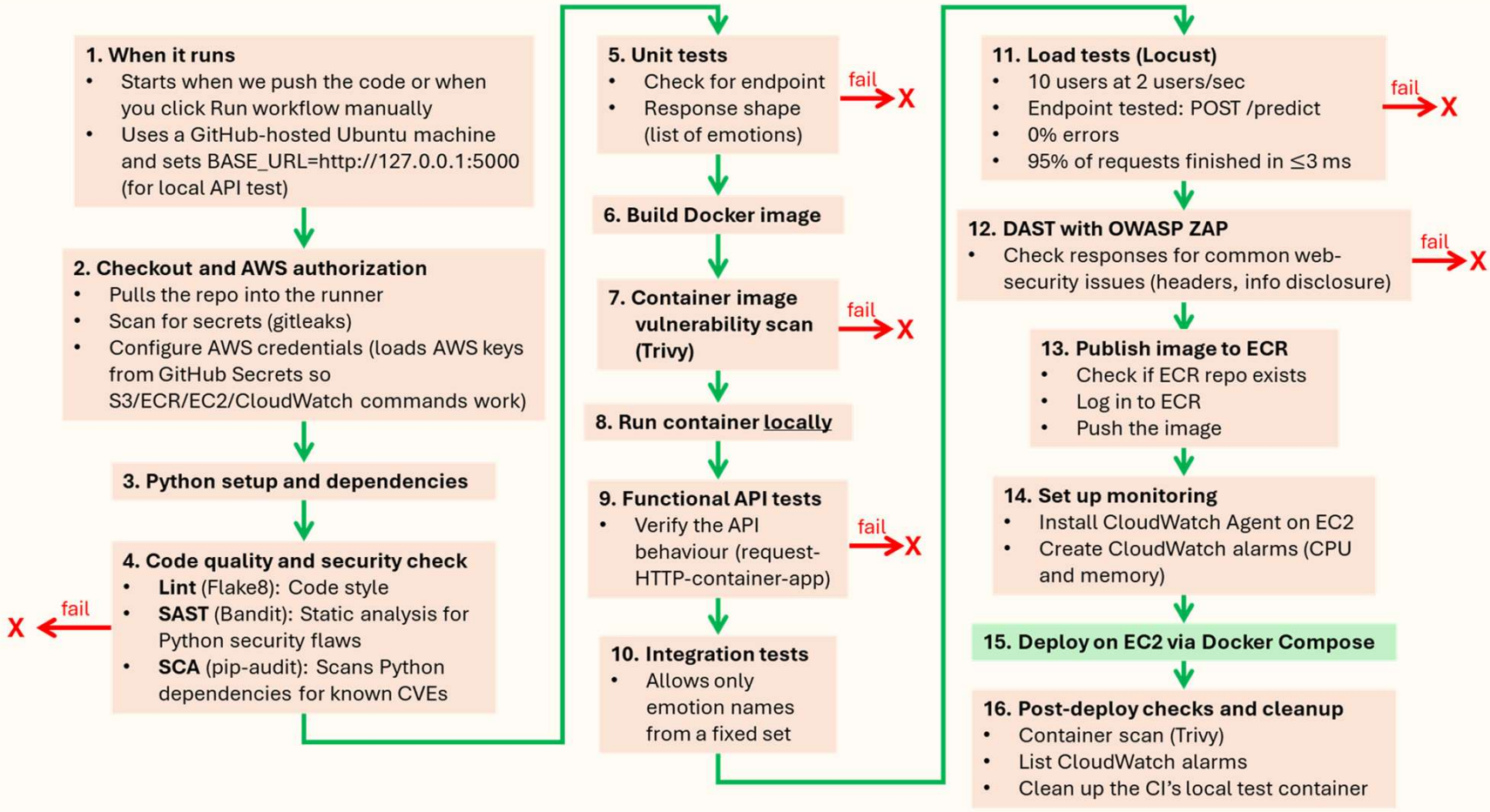


AWS Services

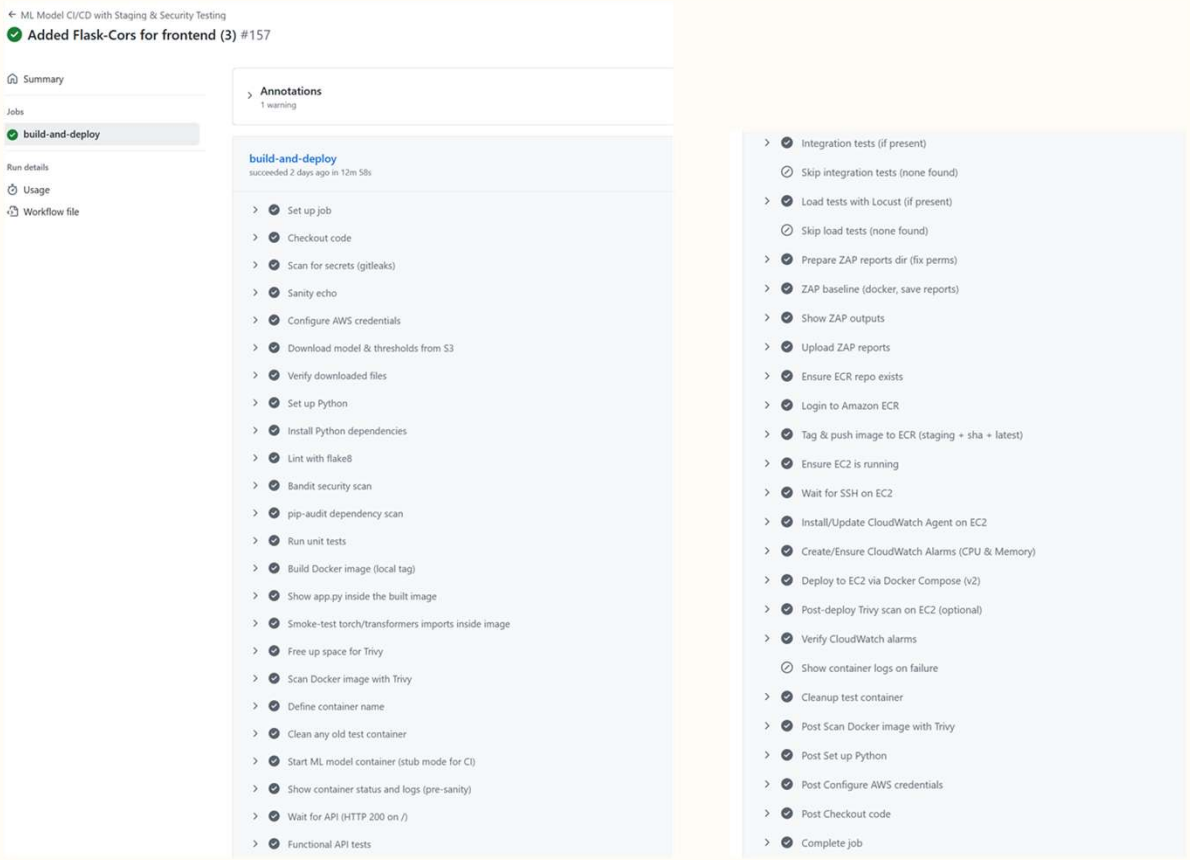
- EC2 Host backend, frontend, and ML model
- S3 Store large ML model files
- RDS Database (MySQL)

- ECR Container image registry
- IAM Role Give EC2 permission to pull from ECR and S3
- CloudWatch Logs and monitoring

Machine Learning Model CI/CD

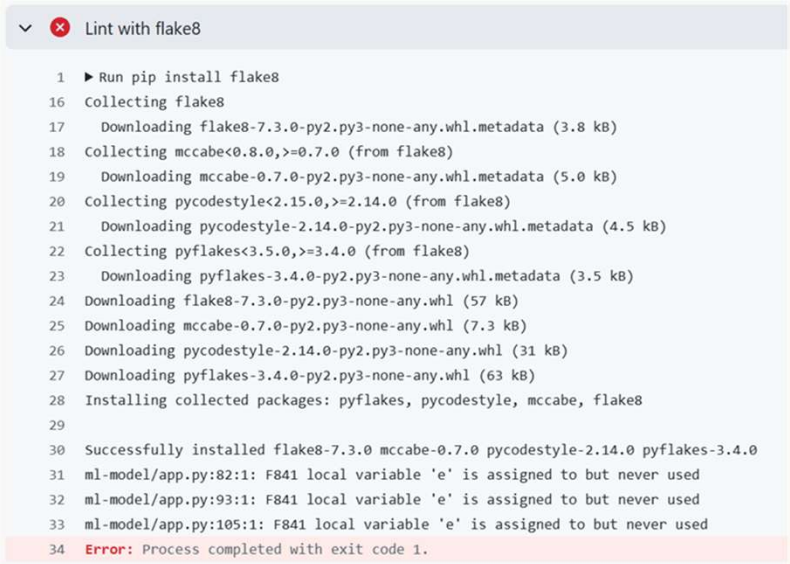


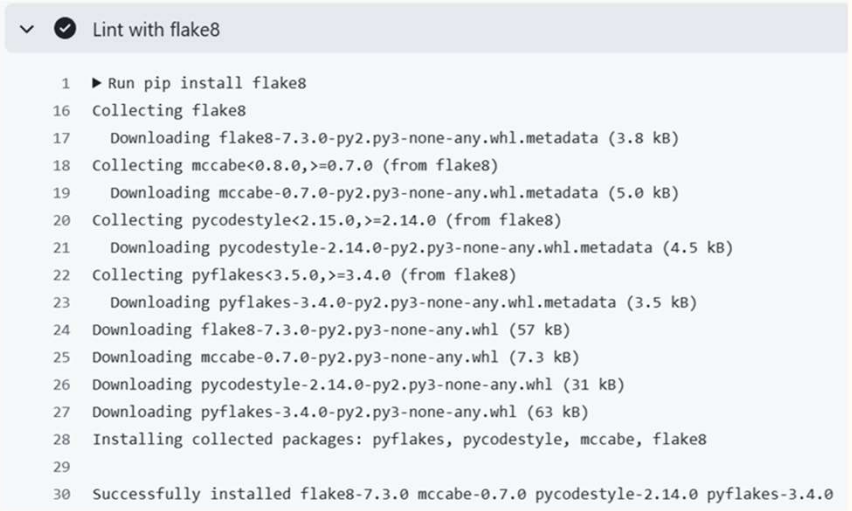
Machine Learning Model CI/CD GitHub Actions



- Note that Terraform and Ansible were not used because for the scale of our project, we do not make use of them very much
- Instead of staging, functional and integration tests were done locally

Machine Learning Model CI/CD
Lint (flake8)





- Lint flagged the local variable that is never used
- Fixed by removing that variable in the code

Machine Learning Model CI/CD
SAST (Bandit)

```
--------------------------------------------------
>> Issue: [B615:huggingface_unsafe_download] Unsafe Hugging Face Hub download without revision pinning in from_pretrained()
   Severity: Medium   Confidence: High
   CWE: CWE-494 (https://cwe.mitre.org/data/definitions/494.html)
   More Info: https://bandit.readthedocs.io/en/1.8.6/plugins/b615_huggingface_unsafe_download.html
   Location: ml-model/ml_model_code.py:168:8
167     # Load DeBERTa model for multi-label classification
168     model = AutoModelForSequenceClassification.from_pretrained(
169         'microsoft/deberta-v3-small', num_labels=8, problem_type='multi_label_classification'
170     )
171

--------------------------------------------------
>> Issue: [B614:pytorch_load] Use of unsafe PyTorch load
   Severity: Medium   Confidence: High
   CWE: CWE-502 (https://cwe.mitre.org/data/definitions/502.html)
   More Info: https://bandit.readthedocs.io/en/1.8.6/plugins/b614_pytorch_load.html
   Location: ml-model/ml_model_code.py:274:26
273     if os.path.exists(best_model_path):
274         model.load_state_dict(torch.load(best_model_path, map_location=device))
275         model.to(device)

--------------------------------------------------
```

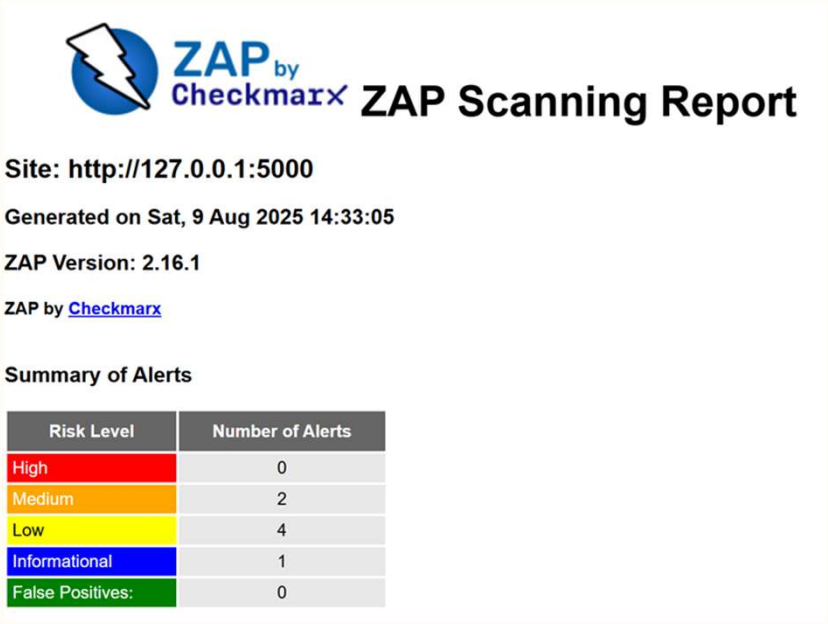
```
--------------------------------------------------

Code scanned:
        Total lines of code: 268
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 7
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 1
                High: 6
Files skipped (0):
Error: Process completed with exit code 1.
```

- SAST scan flagged some insecure coding practices
- It is acceptable to suppress these because
  - The model file is produced by my own training job
  - Stored in AWS S3 with access control and fetched when building the image, not during runtime

Machine Learning Model CI/CD
DAST (OWASP ZAP)



ZAP by Checkmarx **ZAP Scanning Report**

Site: http://127.0.0.1:5000

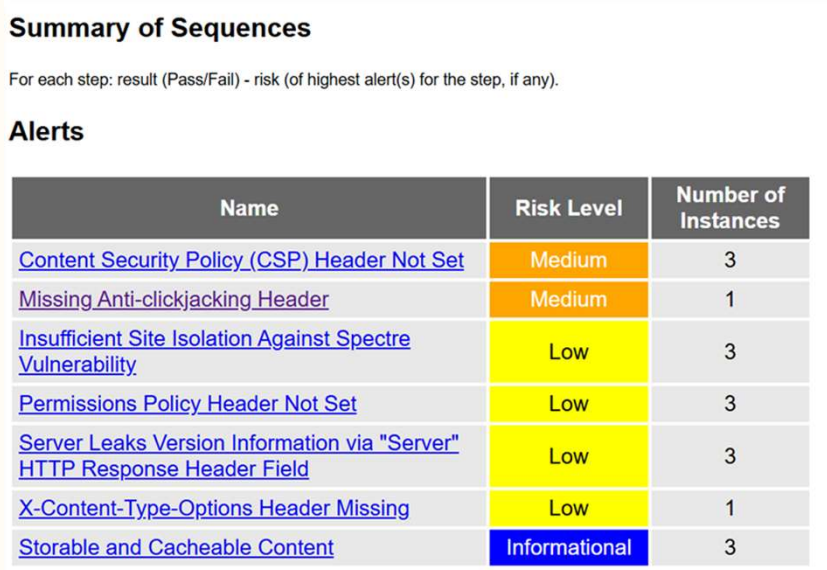
Generated on Sat, 9 Aug 2025 14:33:05

ZAP Version: 2.16.1

ZAP by Checkmarx

Summary of Alerts

Risk Level	Number of Alerts
High	0
Medium	2
Low	4
Informational	1
False Positives:	0



Summary of Sequences

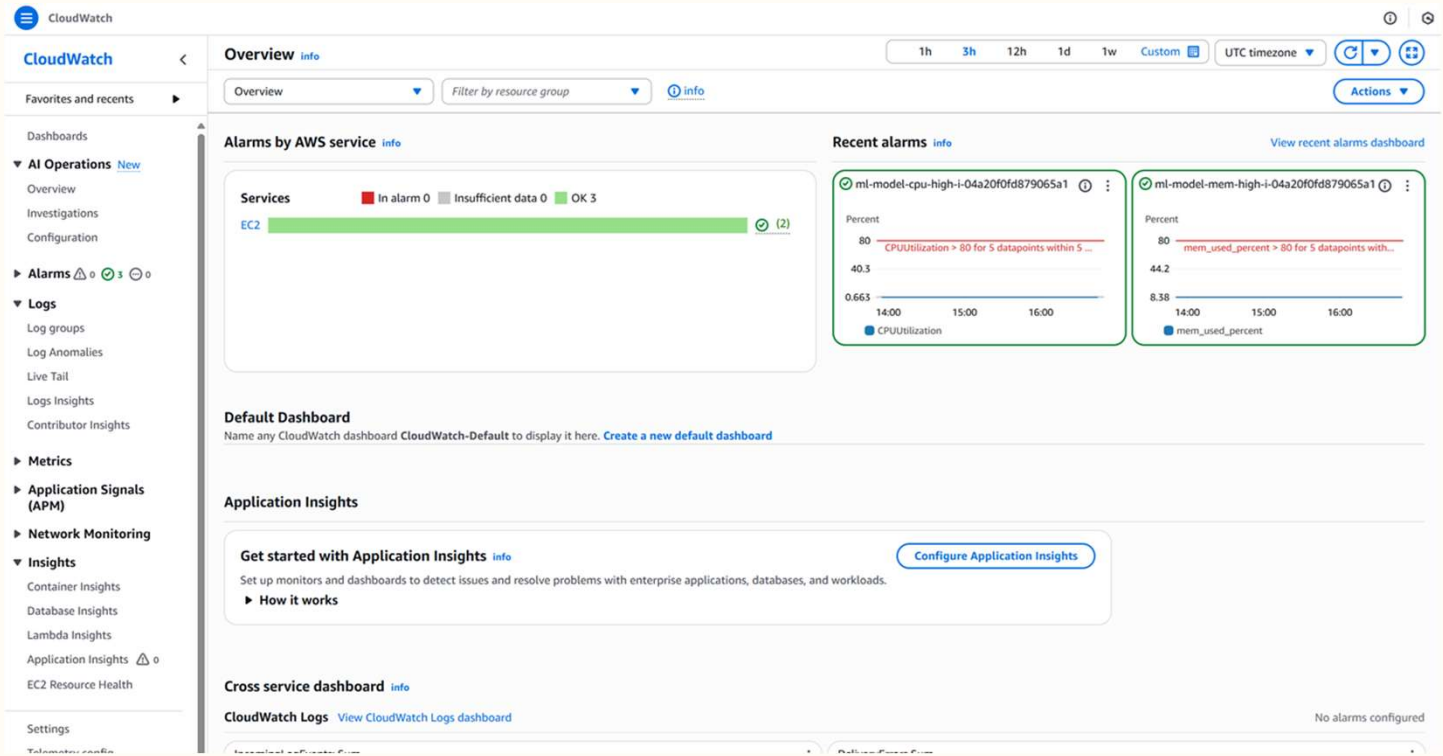
For each step: result (Pass/Fail) - risk (of highest alert(s) for the step, if any).

Alerts

Name	Risk Level	Number of Instances
Content Security Policy (CSP) Header Not Set	Medium	3
Missing Anti-clickjacking Header	Medium	1
Insufficient Site Isolation Against Spectre Vulnerability	Low	3
Permissions Policy Header Not Set	Low	3
Server Leaks Version Information via "Server" HTTP Response Header Field	Low	3
X-Content-Type-Options Header Missing	Low	1
Storable and Cacheable Content	Informational	3

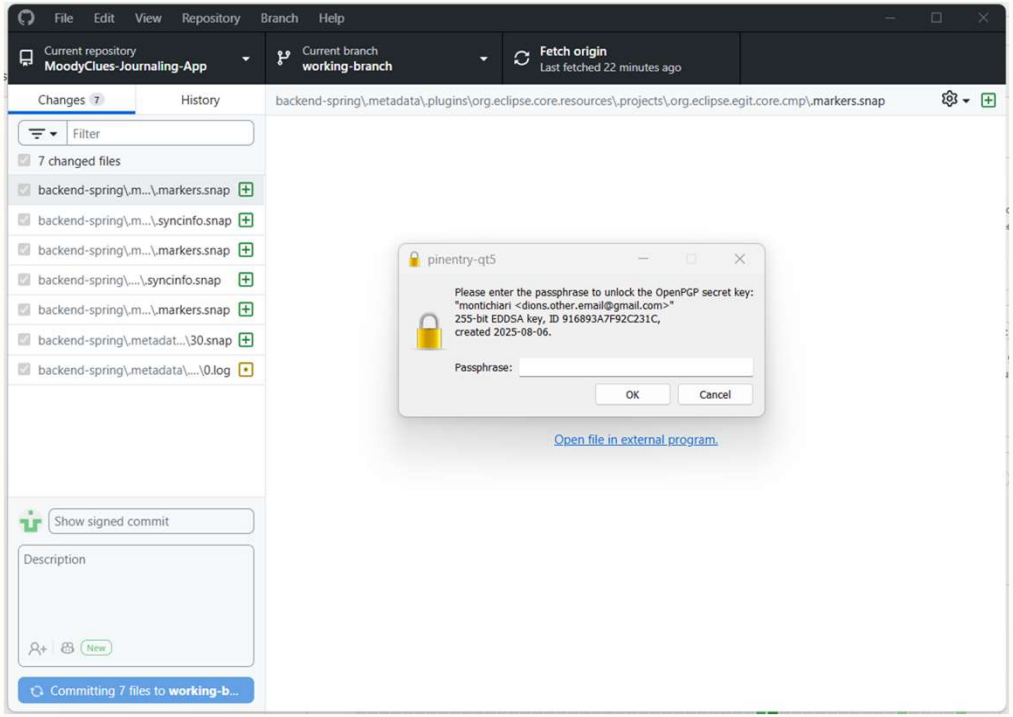
- DAST scan showed some alerts, but none were high risk

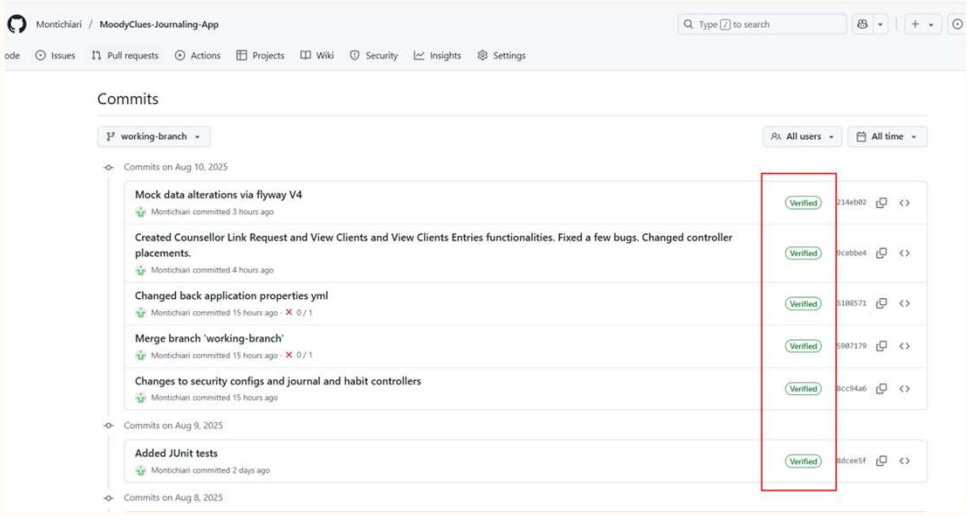
Machine Learning Model CI/CD
AWS CloudWatch



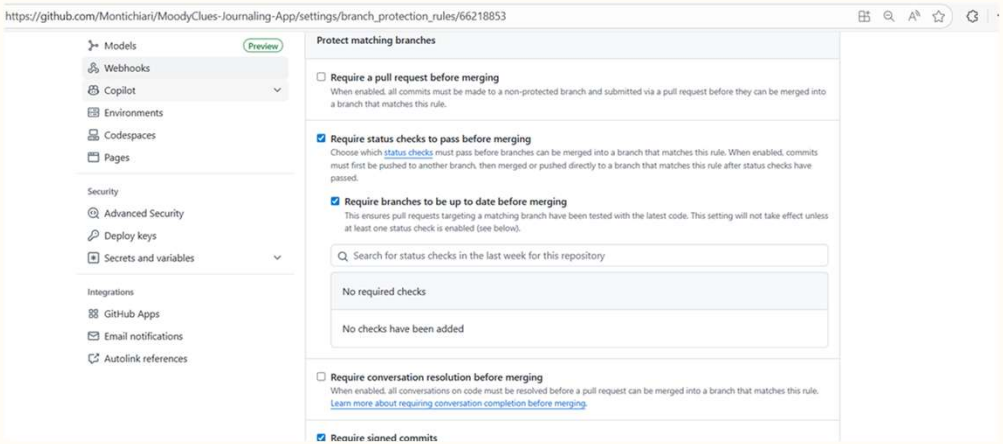
- Set up CloudWatch for post-deployment monitoring
- Alerts when CPU or memory utilization exceeds 80%

Backend CI/CD
GitHub Signed Commits

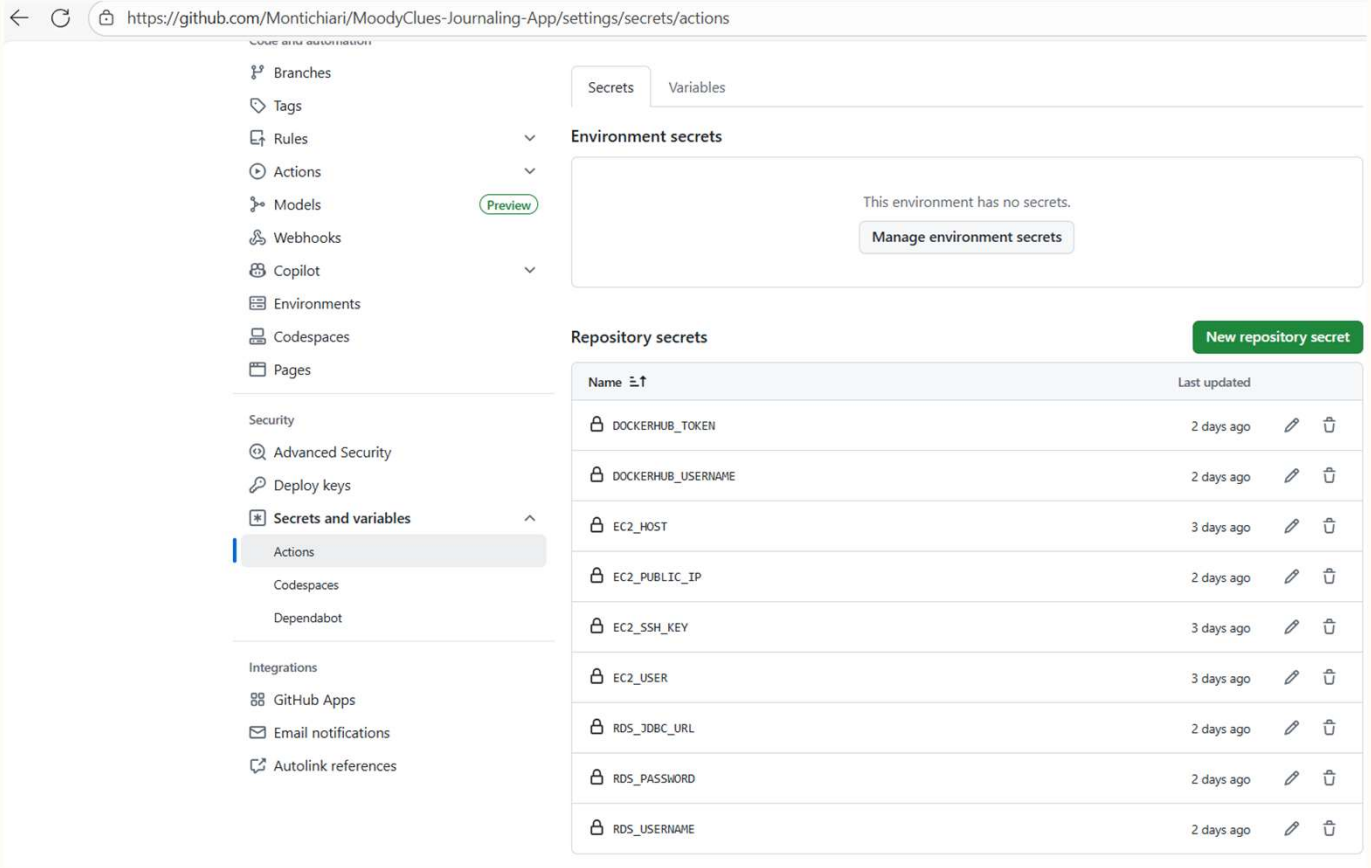




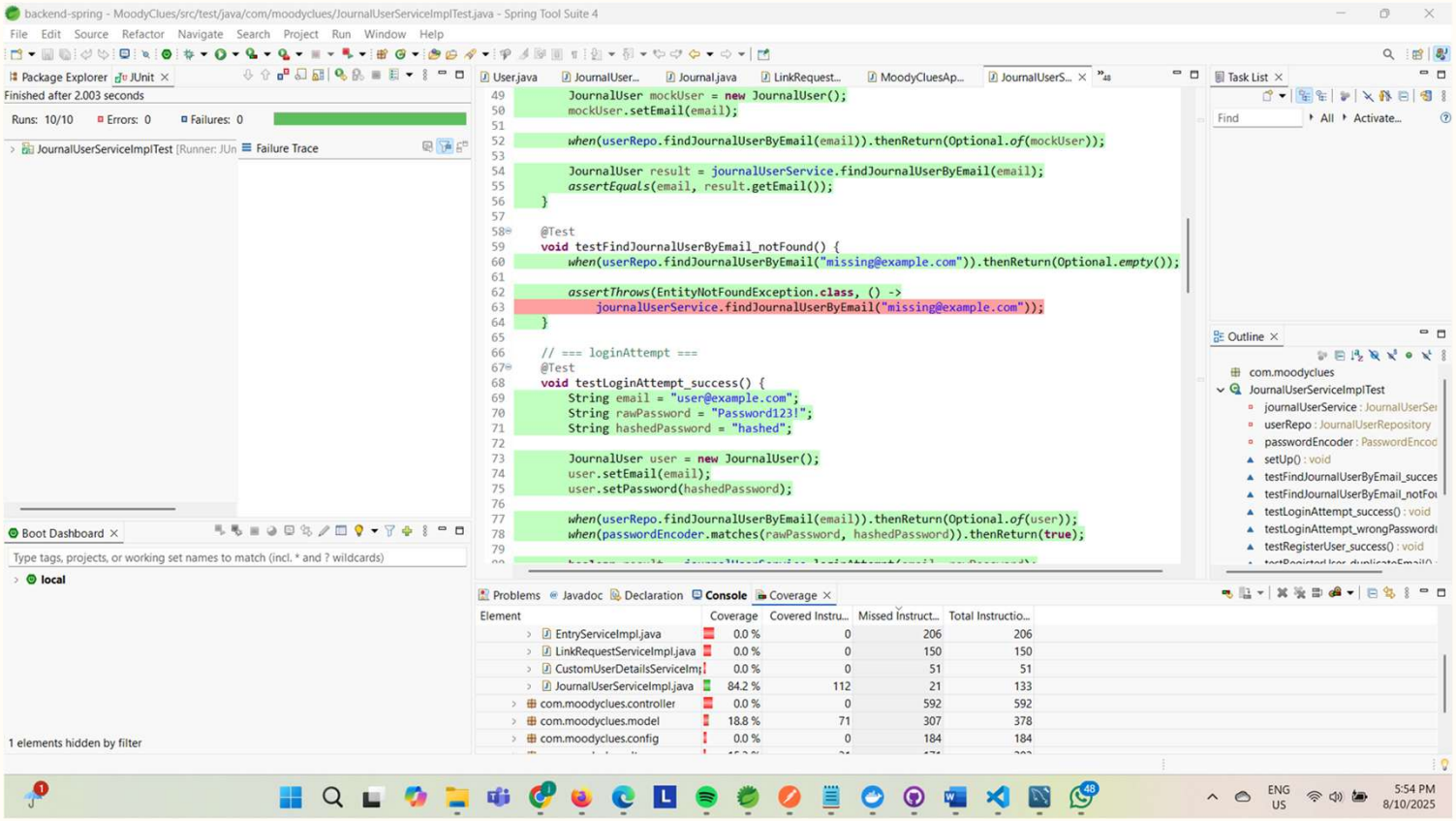
Branch Protection



Backend CI/CD Secrets Protection



Backend CI/CD
JUnit Tests + Code coverage (JaCoCo)



Backend CI/CD
GitHub Actions Pipeline

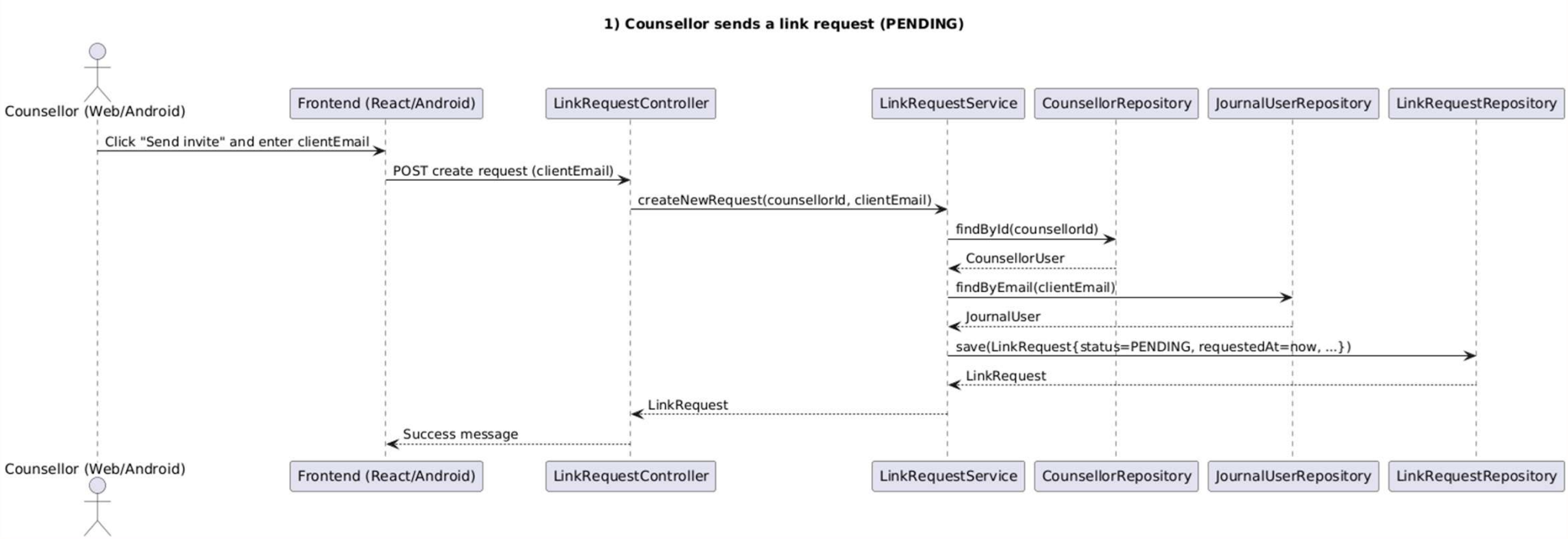
GitHub / Actions	Flyway Data Migration	Build, Dockerize, Push	Deploy on EC2	Success
Upon signed commit and push to Main branch, starts the process by taking the code from the “backend-spring” folder. Ensures only 1 deploy is run at a time.	GitHub Actions connects directly to database on RDS and runs Flyway. Flyway: Scans migration folder for versioned scripts -> Checks/creates flyway_schema_history -> Applies only the pending migrations.	The JAR is built, Docker logged into with secrets, image created and pushed to DockerHub.	The script then SSH into the EC2 instance, stops any running container instance, pulls latest image from DockerHub, and runs the image.	The backend is now updated with the latest changes / features automatically.

Backend CI/CD
SAST

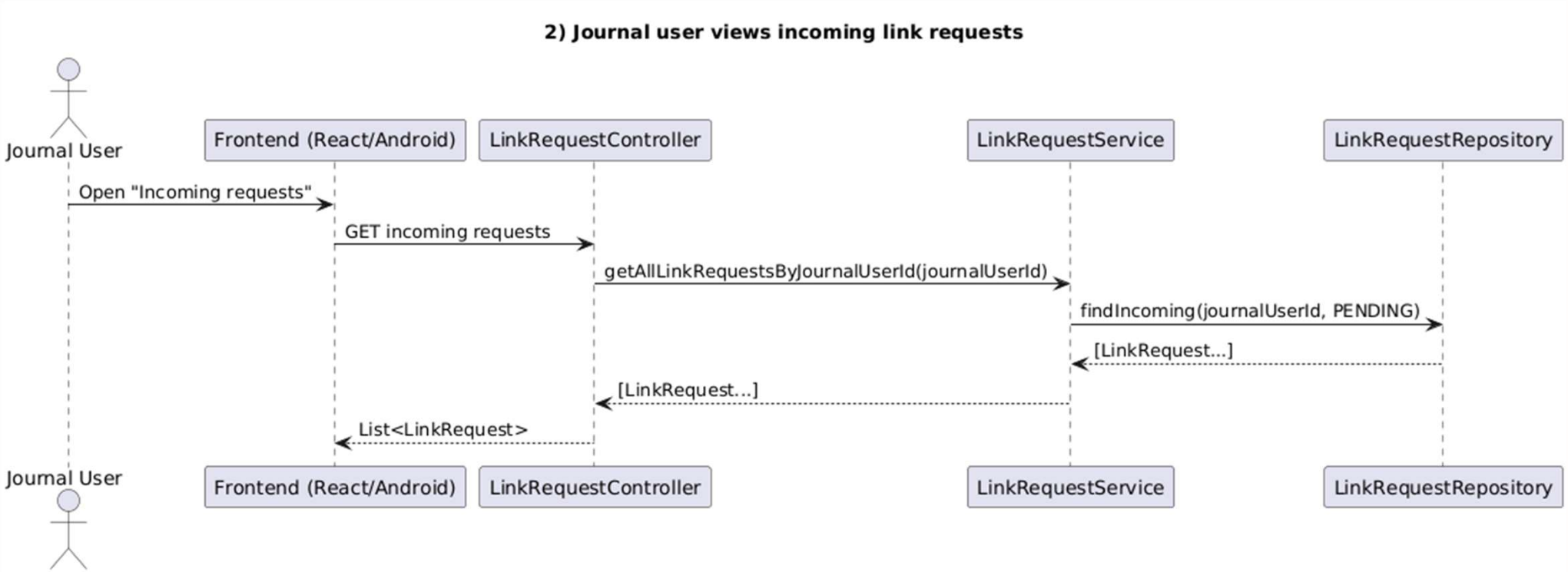
Selected Features

Feature Highlight #1 - Link Requests

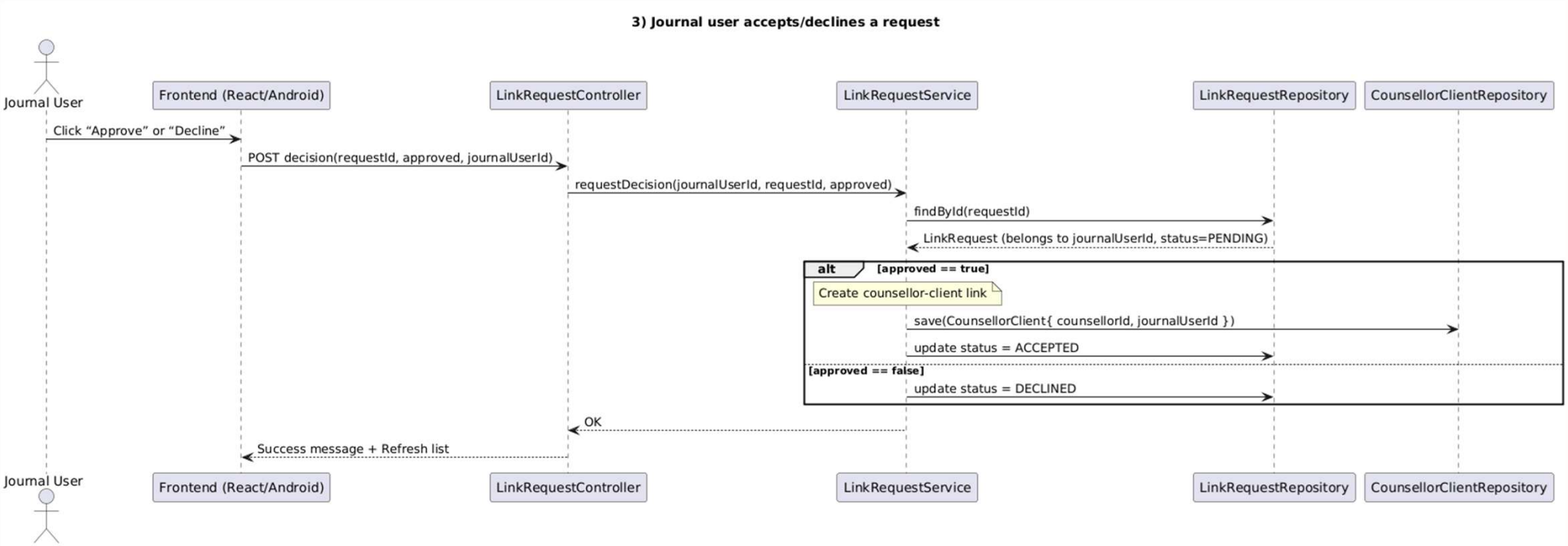
Core feature needed to link Counsellor with JournalUser, to allow them to access their clients' entries, and have their projections in dashboard.



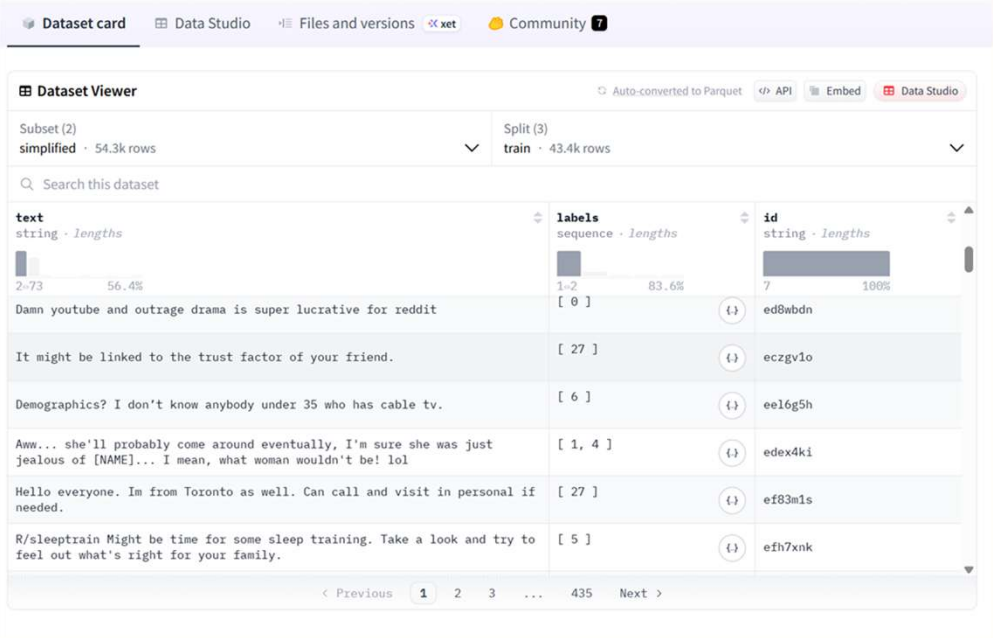
Feature Highlight #1 - Link Requests

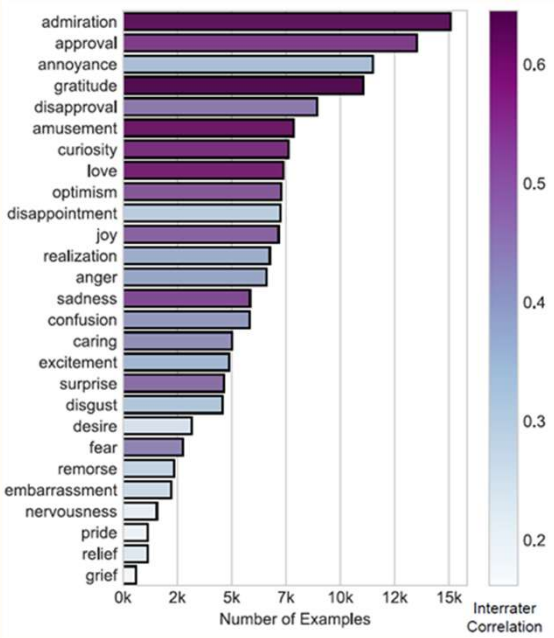


Feature Highlight #1 - Link Requests



Feature Highlight #2 - Machine Learning
Machine Learning Model for Sentiment Analysis





- Used GoEmotins dataset (Google Research)
- Number of examples: 58,009
- Number of emotions: 27 + neutral

Dorottya Demszky, Dana Movshovitz-Attias, Jeongwoo Ko, Alan Cowen, Gaurav Nemade, and Sujith Ravi. 2020. GoEmotions: A Dataset of Fine-Grained Emotions. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4040–4054, Online. Association for Computational Linguistics. (10.18653/v1/2020.acl-main.372)

Feature Highlight #2 - Machine Learning Remapping the Emotions

Happy

- Admiration
- Amusement
- Approval
- Caring
- Excitement
- Gratitude
- Joy
- Love
- Optimism
- Pride
- Relief

Sad

- Disappointment
- Grief
- Remorse
- Sadness

Angry

- Anger
- Annoyance
- Disapproval
- Disgust

Anxious

- Fear
- Nervousness
- Embarrassment

Curious

- Curiosity
- Desire

Surprised

- Surprise

Confused

- Confusion
- Realization

Neutral

- 27 emotions were regrouped into 7 emotions with neutral to capture the basic emotions and decrease the model size
- 7 was chosen because if more than 7, there were many emotion groups with one or two emotions, and the number of samples in these emotion groups are very small (imbalanced data)

Feature Highlight #2 - Machine Learning Model Training

- **Converted to multi-hot vector**
  - Emotion_classes = ['angry', 'sad', 'anxious', 'happy', 'curious', 'confused, 'surprised', 'neutral']
  - E.g. if the emotions are 'happy and 'anxious': Multi-hot vector: [0, 0, 1, 1, 0, 0, 0, 0]
- **Tokenization**
  - Used DeBERTa tokenizer and tokenized all training, validation, and test texts with truncation at a maximum length of 64 tokens (4-6 sentences)
  - DeBERTa: transformer based model developed by Microsoft, widely used for a variety of NLP tasks
- **Class weights calculation**
  - Computed class weights to address class imbalance using the ratio of the samples to total samples
- **Model definition**
  - Used the pre-trained DeBERTa model for multi-label classification
  - 8-class output (7 emotions + neutral)
- **Training loop**
  - AdamW optimizer with a learning rate of 2.5e-5 (optimized with small dataset)
  - Batch size = 16 (optimized with small dataset)
  - Epochs = 10, implemented early stopping based on macro F1 score improvement
  - Evaluated model performance on the validation set using fixed thresholds for multilabel predictions (threshold=0.5)
- **Threshold tuning**
  - Tuned per-class thresholds on the validation set to maximize F1 scores using a range of thresholds
  - Saved the best model based on F1 score
- **Fallback logic**
  - If no emotion exceeds the threshold, assign the top predicted emotion

Feature Highlight #2 - Machine Learning
ML Model Evaluation

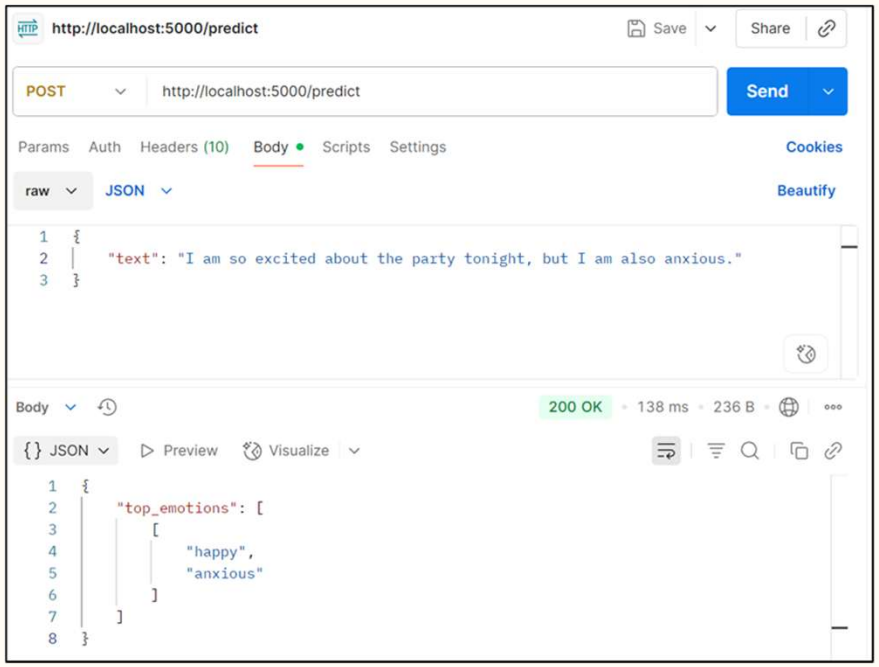
Evaluation on test set	
F1 Score (Micro)	0.658
F1 Score (Macro)	0.574
Hamming Accuracy	**0.900**

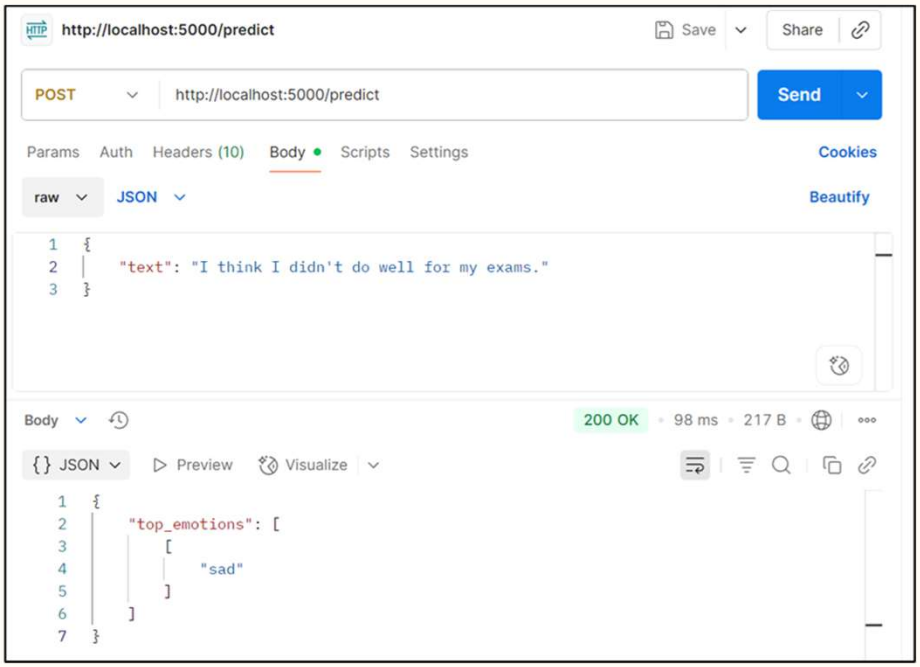
- For multi-label classification tasks, F1 scores of 0.6-0.7 (micro average) is typical, especially when the classes are imbalanced
- Hamming accuracy of 0.9 is strong, indicating that the model is correctly predicting most of the labels per instance

	Precision	Recall	F1 Score	Support
Angry	0.54	0.65	0.59	780
Sad	0.52	0.61	0.56	331
Anxious	0.53	0.64	0.58	128
Happy	**0.77**	**0.84**	**0.80**	1966
Curious	0.43	0.68	0.53	338
Confused	0.31	0.40	0.35	275
Surprised	0.51	0.57	0.54	136
Neutral	0.58	0.74	0.65	1787
Micro avg	0.60	0.73	0.66	5741
Macro avg	0.52	0.64	0.57	5741

- The macro F1-score is lower than micro because certain labels are much easier to predict (happy, neutral) than others (confused, curious)

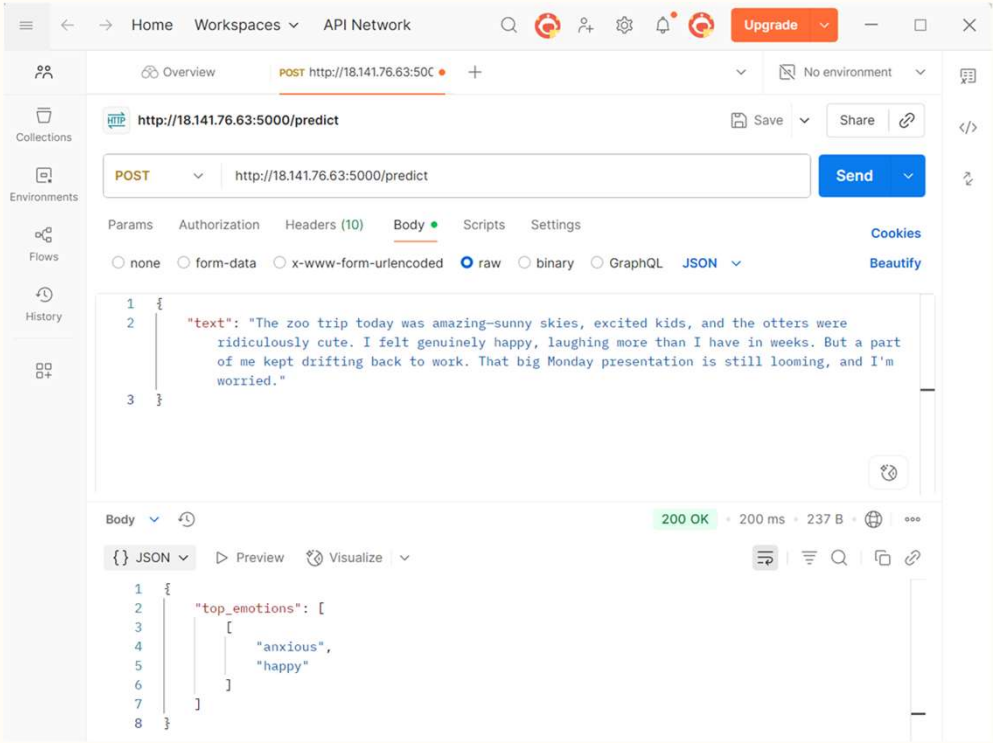
Feature Highlight #2 - Machine Learning Checking the model

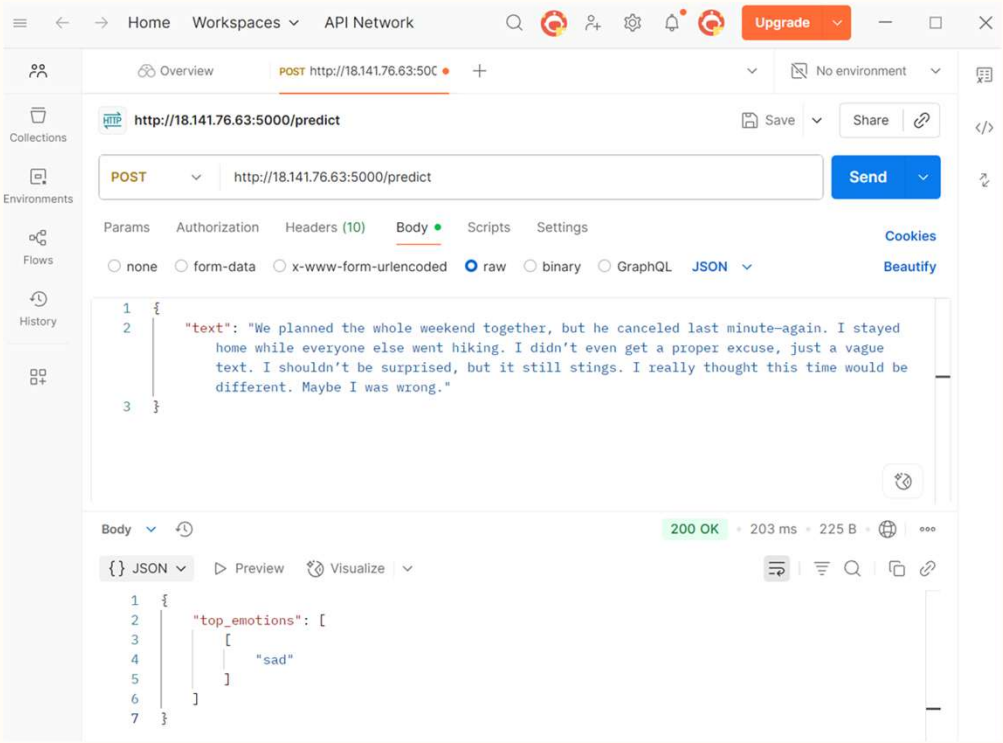




- The predictions of the emotions are reasonable

Feature Highlight #2 - Machine Learning
Checking the model with longer text





- The text above are close to 64 tokens (maximum length)
- The predictions of the emotions are reasonable

Feature Highlight #3 - Dashboards

The User Dashboard required 3 projections of data:

1. Journal data averaged by day
1. Total count of emotions
1. Habits data by day

The Counsellor Dashboard doesn't need Habits, but needs the "Clients" they are linked with.
On the frontend, during render, a data hook is used to call the API from the backend, then sets the data in state. The parent dashboard page pulls from the hook, and passes as props to the various child components. The component with the date picker display the value via prop, and onChange, calls the parent to setState, and re-render

Appendix

Problem statement sources:

https://www.imh.com.sg/Newsroom/News-Releases/Documents/NYMHS_Press%20Release_FINAL19Sep2024.pdf

https://www.channelnewsasia.com/singapore/poor-mental-health-young-adults-seek-help-moh-survey-3802531

Efficacy of journaling in the management of mental illness: a systematic review and meta-analysis